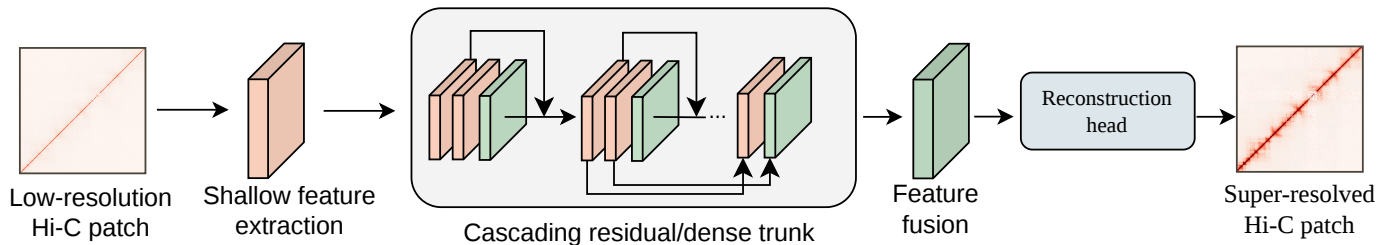
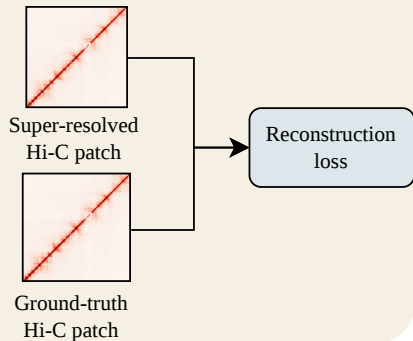


(A) Super resolution pipeline

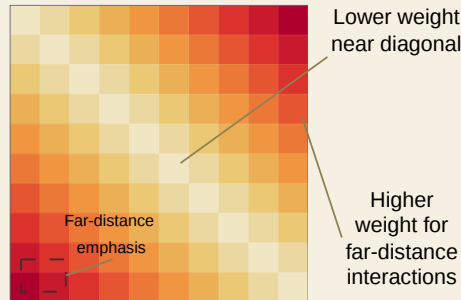


(B) Distance-aware loss design

(B1) Reconstruction supervision



(B2) Distance-aware weighting



(C) Final optimization objective

$$L_{\text{total}} = L_{\text{rec}} + \lambda_{\text{dist}} L_{\text{dist}}$$

$$L_{\text{rec}} = \|\hat{Y} - Y\|_1$$

$$L_{\text{dist}} = \sum_{i,j} w_{ij} |\hat{Y}_{ij} - Y_{ij}|$$

w_{ij} increases with genomic distance;
DistFarStrong further emphasizes long-range contacts.